ECE 1305 – Introduction to C Programming
Tutorial 04
Memory Map of Floating-Point Numbers

Write a C++ program that declares 5 type **`double`** variables and assigns a value to those variables. Have the program create a Memory Map with the following form using the **`"\t"`** operator to create a "tab" spacing. Determine the spacing between adjacent memory locations.

Input

```
// Tutorial 04 - Memory Map but with Floating Point Numbers
#include <iostream>
using namespace std;

int main()
{
    double a = 1.1, b = 2.2, c = 3.3, d = 4.4, e = 5.5;

    cout << "Variable\tValue\tAddress\n";
    cout << "a\t\t" << a << "\t" << &a << endl;
    cout << "b\t\t" << b << "\t" << &b << endl;
    cout << "c\t\t" << c << "\t" << &c << endl;
    cout << "d\t\t" << d << "\t" << &d << endl;
    cout << "e\t\t" << e << "\t" << &e << endl;

    cout << "\nMemory Spacing:\n";
    cout << ((&b - &a) * sizeof(double)) << " bytes\n";
    cout << ((&c - &b) * sizeof(double)) << " bytes\n";
    cout << ((&d - &c) * sizeof(double)) << " bytes\n";
    cout << ((&e - &d) * sizeof(double)) << " bytes\n";

    system("pause");
    return 0;
}
```

Output

```
C:\Users\anmal\Desktop\ECE
Variable        Value   Address
a               1.1     00000098BD74FA58
b               2.2     00000098BD74FA78
c               3.3     00000098BD74FA98
d               4.4     00000098BD74FAB8
e               5.5     00000098BD74FAD8

Memory Spacing:
32 bytes
32 bytes
32 bytes
32 bytes
Press any key to continue . . .
```